

Editorial

by Emi Tanaka

In this issue

On behalf of the Editorial Board, I am pleased to present Volume 18, Issue 3 of the R Journal. This issue features twenty research articles spanning a wide range of developments in statistical methodology, machine learning, time series, spatial analysis, causal inference, visualisation, and scientific computing.

Several articles introduce new tools for computationally challenging statistical problems. **hdtg** provides methods for simulating from high-dimensional truncated multivariate normal distributions, while **ODRF** develops an implementation of oblique decision trees and random forests. **psvmSDR** provides tools for sufficient dimension reduction using principal machines, and **DRIP** brings jump regression methods to image analysis. **CRANpkg[SelectionBias]** implements a range of methods for bounding selection bias in causal estimands. **LBBNN** provides a scalable approach to sparse and interpretable Bayesian neural networks, making use of variational inference and GPU acceleration. The issue also includes **rSRD** supporting the sum of ranking differences procedure, and **xplainfi** brings together a broad range of feature importance methods with tools for statistical inference.

Time series modelling is also well represented in this issue. **mtarm** provides Bayesian estimation and forecasting for multivariate threshold autoregressive models, while **echos** introduces automated time series forecasting using echo state networks. **MSTest** provides a collection of procedures for testing the number of regimes in Markov-switching models, **VARshrink** brings together shrinkage methods for high-dimensional vector autoregressive models, and **iAR** provides tools for modelling irregularly observed time series. **demofit** brings parametric and stochastic approaches to mortality modelling and forecasting into a unified framework.

There are also several contributions in spatial, spatiotemporal, and multivariate analysis. **ldmppr** provides tools for modelling marked point processes where the marks may depend on location, while **sugarglider** extends glyph-map visualisation for exploring spatiotemporal data. **goSorensen** provides methods for comparing gene lists using Gene Ontology information, **admix** supports estimation, testing and clustering for admixture models, and **dbrobust** provides tools for robust distance-based analysis and visualisation of mixed-type data.

Finally, **metafrontier** provides a comprehensive set of tools for metafrontier analysis of efficiency and productivity.

All packages discussed are available on CRAN. Supplementary material with fully reproducible code is available for download from the R Journal website.

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